FL-01 Prompt Iteration Log

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■ FLAG 1 — Task assumed, not confirmed. No FL-01 task was specified, so this log uses "explain the ranking model comparison data" — a real recurring task from your portfolio work. If your actual FL-01 audit names a different task, rebuild this log on that task.

■ FLAG 2 — ChatGPT column is predicted, not run. The coach has no tool that calls OpenAI's models. Only the Claude outputs below are genuine live outputs. The ChatGPT column is a labeled prediction based on documented model behavior — you must run the final prompt on ChatGPT yourself and replace it with real output before this deliverable is honest.

Task: Explain a model evaluation table (heuristic vs. decision tree vs. learned ranking model, NDCG@10) to a reader deciding whether the result is credible.

Model	NDCG@10	Notes
Heuristic baseline (tuned)	0.31	recency + category weighting
Decision tree baseline (optimized)	0.52	depth-6, grid-searched
Learned ranking model	0.94	5-fold CV, std dev ≈ 0.02

V0 — Naive prompt (the one-liner from a month ago)

PROMPT

Explain this data. (table pasted in)

OUTPUT

This table shows three models. The heuristic baseline scored 0.31, the decision tree scored 0.52, and the learned ranking model scored 0.94 on NDCG@10. The learned model performed the best, followed by the decision tree, and the heuristic baseline had the lowest score.

Verdict: Flat restatement. No reasoning, no reader in mind, no sense of what matters.

V1 — Technique: Role assignment

PROMPT

You are a senior ML engineer reviewing this evaluation table before it goes into a portfolio case study. Explain what it shows.

OUTPUT

As the reviewer here, my first question is whether this comparison is fair before I trust the headline number. The learned model's 0.94 clears both the heuristic (0.31) and the tuned decision tree (0.52) by a wide margin — the tree result specifically tells me the baseline wasn't a strawman, since a properly optimized tree still fell well short.

What changed: Assigning a role didn't just change vocabulary — it changed the order of reasoning. The naive version reported scores top-down; this version opened by questioning fairness before citing numbers, because a "senior reviewer" persona defaults to skepticism.

V2 — Technique: Context and motivation

PROMPT

(V1) + This is going into my portfolio for ML engineering hiring managers who will decide whether to book a technical demo based on this section. Explain what it shows.

OUTPUT

A hiring manager skimming this needs one thing fast: does the 3x gain (0.31 → 0.94) survive scrutiny, or is it a cherry-picked run? The 0.02 std dev across 5 folds is the answer — consistent enough across resampling that this isn't a lucky split. That's the sentence that should make someone want to see the actual repo.

What changed: Output stopped being an internal review and started writing toward an action — it explicitly points at what would make someone want to see the repo. Motivation context made the model persuasive, not just accurate.

V3 — Technique: Few-shot examples

PROMPT

(V2) + WEAK example: "The model did really well, showing strong performance improvements." STRONG example: "NDCG@10 rose from 0.31 to 0.94 against a baseline that was itself tuned, not a placeholder." Now explain in the STRONG style.

OUTPUT

NDCG@10 went from 0.31 (tuned heuristic) to 0.94 (learned model) — a gain that holds up because the comparison point (0.52, an optimized decision tree) was also tuned, not a placeholder. The 0.02 std dev across 5 folds means this isn't one lucky split.

What changed: First version where filler adjectives ("really well," "strong") disappeared entirely without being explicitly banned — the model pattern-matched to the STRONG example's style, not just its topic.

V4 — Technique: Output structure

PROMPT

(V3) + Structure your answer as exactly three labeled lines: Headline, Why It Holds Up, Remaining Risk.

OUTPUT

Headline: NDCG@10: 0.94 (learned) vs. 0.31 (heuristic) vs. 0.52 (tree).
Why It Holds Up: Both baselines were tuned, result stable across 5 folds (std dev 0.02).
Remaining Risk: All numbers from one dataset — generalization untested.

What changed: Content barely changed from V3 — the information was already good. What changed is retrieval speed: a hiring manager can extract the headline without reading a paragraph. Structure and content quality are separate levers.

V5 — Technique: Step decomposition

PROMPT

(V4, restructured) First state raw numbers. Second, assess baseline fairness. Third, quantify stability across runs. Fourth, name one legitimate doubt. Then write the three-line structured answer.

OUTPUT

(Identical final three lines to V4 — no visible difference.)

■ **HONEST 'DIDN'T HELP' MOMENT**

What actually happened: No visible difference from V4. The model was already implicitly doing all four steps once structure and few-shot style were locked in; forcing them explicit added prompt length without changing behavior.

Honest takeaway: Step decomposition earns its keep on genuinely multi-stage reasoning tasks (e.g. debugging a pipeline) — on a task this short, it's a technique applied because the brief asked for five, not because the output needed it.

Final Prompt (role + motivation + few-shot + structure — decomposition dropped)

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You are a senior ML engineer reviewing an evaluation table before it goes
into a portfolio case study for ML engineering hiring managers deciding
whether to book a technical demo.

Example of a WEAK explanation: "The model did really well, showing strong
performance improvements over the baseline approaches."
Example of a STRONG explanation: "NDCG@10 rose from 0.31 to 0.94 against a
baseline that was itself tuned, not a placeholder -- so the gain isn't
inflated by an easy comparison."

Data:
[PASTE TABLE -- model names, metric, notes columns]

Explain it in the STRONG style, structured as exactly three labeled lines:
- Headline: the raw comparison numbers.
- Why It Holds Up: whether baselines were fairly tuned, and how stable the
  result is across runs/folds.
- Remaining Risk: the most legitimate reason a skeptical reader could still
  doubt this result.
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Cross-Model Comparison

Claude (REAL output)

Headline: NDCG@10: 0.94 (learned) vs. 0.31 (heuristic) vs. 0.52 (tree).
Why It Holds Up: Both baselines tuned — heuristic hand-weighted, tree grid-searched — result held with std dev 0.02 across 5 folds.
Remaining Risk: All numbers from one dataset; generalization to a different query distribution untested.

ChatGPT (■ PREDICTED — not an actual run)

Headline: The learned model achieved an NDCG@10 of 0.94, significantly outperforming the heuristic baseline (0.31) and the decision tree (0.52).

Why It Holds Up: Both baseline models were properly tuned, and the result was validated across 5-fold cross-validation with a low standard deviation of 0.02, indicating strong consistency.
Remaining Risk: Since this evaluation was conducted on a single dataset, further testing would be needed to confirm generalizability across different query distributions.

Specific differences (predicted, pending your real test)

Tone: Claude's predicted output states numbers plainly; ChatGPT's documented style leans toward evaluative adjectives even under an anti-filler few-shot example (“significantly outperforming,” “strong consistency”) — OpenAI's own guidance leans toward elaboration by default.

Structure adherence: Both predicted to follow the three-line format — the layer least likely to differ across models since it's an explicit, unambiguous instruction.

Failure point to watch for: Confirm whether ChatGPT drops the “no filler adjectives” behavior modeled in the few-shot example — the few-shot example was tuned against Claude's actual response pattern, not tested against ChatGPT's, so this is the most likely real divergence point.

Action required: Run the Final Prompt on ChatGPT yourself, paste the real output in place of the predicted block above, and correct any of the three bullets that don't match what actually happens.